



UNIVERSITAS
GADJAH MADA

Natural Infiltration Potential Identification of Seputih Watershed

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Introduction

- Water resources play an **important role** in human life, and it undergoes hydrological processes in a **watershed** that are continuous with each other (Lal, 2000).
- One of the ways to **manage a watershed** is by conducting an **analysis** related to the **potential for natural infiltration** using several parameters
- **Way Seputih watershed** is one of the **priority watersheds** for management by the Lampung Government
- Moreover, it is interesting to identify the infiltration potential of Way Seputih, which facilitate by **remote sensing and GIS Technology**.

Dua Hari Terisolasi: Warga Cabang Lampung Tengah Terpinggirkan di Tengah Banjir



Haluan
4 Maret 2024 14 views



Tujuh Kabupaten di Lampung Tergenang Banjir, Truk dan 4 Rumah Hanyut

Jumat, 10 Maret 2023 - 08:20 WIB

937

Tiga kampung di Lampung Tengah terdampak banjir akibat hujan deras

© Kamis, 9 Maret 2023 10:25 WIB



Salah satu kampung di Kecamatan Terusan Nyunyal Kabupaten Lampung Tengah yang erdampak banjir akibat hujan deras yang melanda daerah ini pada Sabtu (4/3/2023).



Study Area

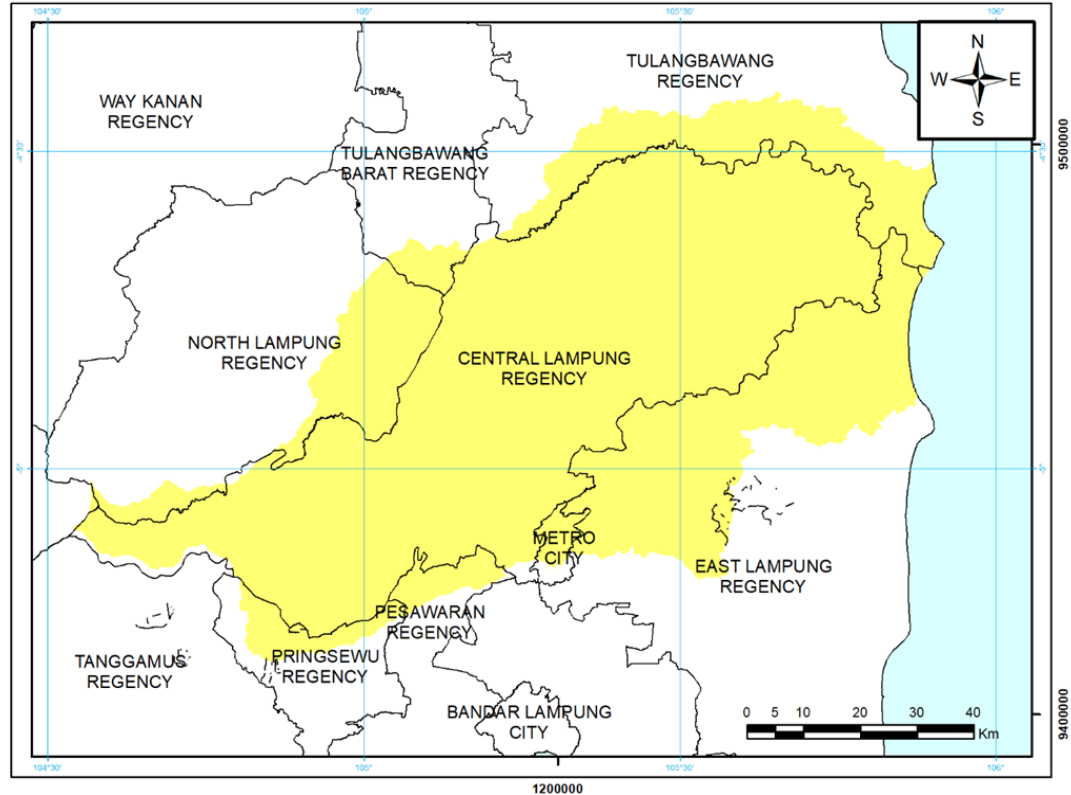
Indonesia



Sumatera



- Regency Boundary
- Seputih Watershed



Tools & Materials

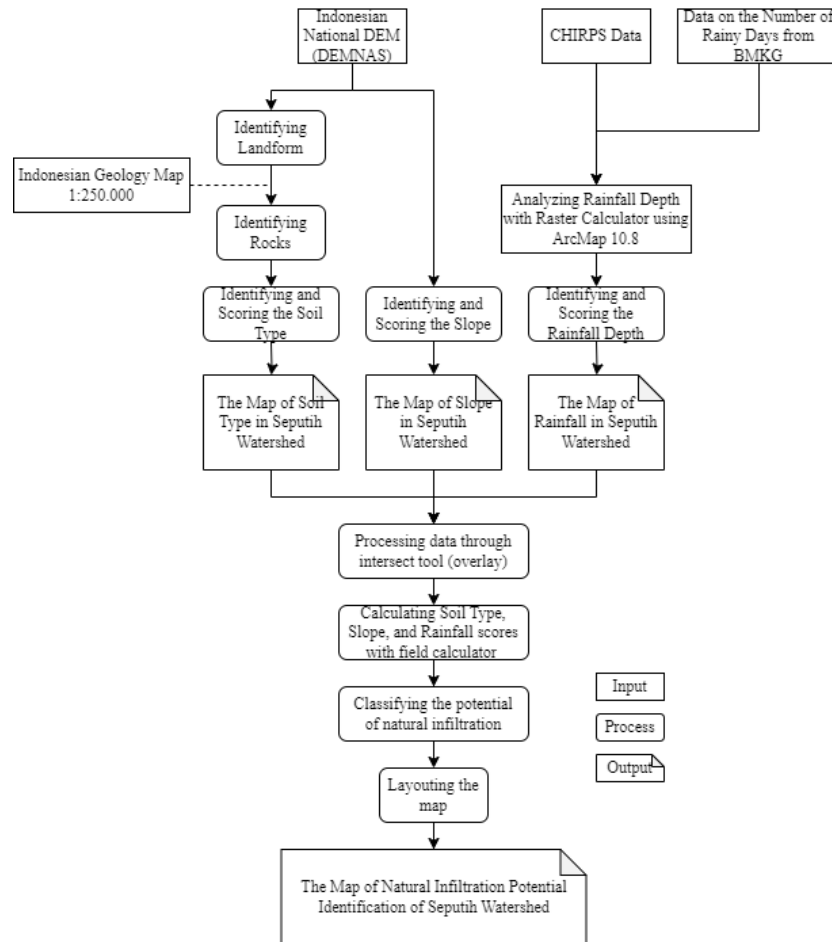
Tools:

1. **ArcMap 10.8** to process the geospatial data
2. **Microsoft Office** used to write the narrative analysis of our research.

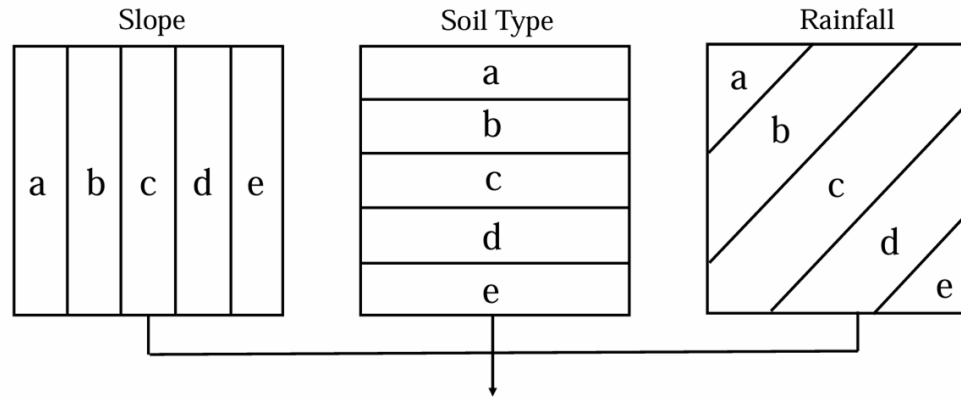
Materials:

1. **Watershed Boundary** Data for analyze the location of Seputih Watershed
2. **CHIRPS data** annual rainfall amount for the year 2023
3. **DEMNAS** with a spatial resolution of 30 meters used to analyze the slope and soil classification
4. **Indonesian Geology Map** 1:250.000 of Lampung
5. **Data on Number of Rainy Days** of BMKG for determine daily rainfall

Data Analysis



Method

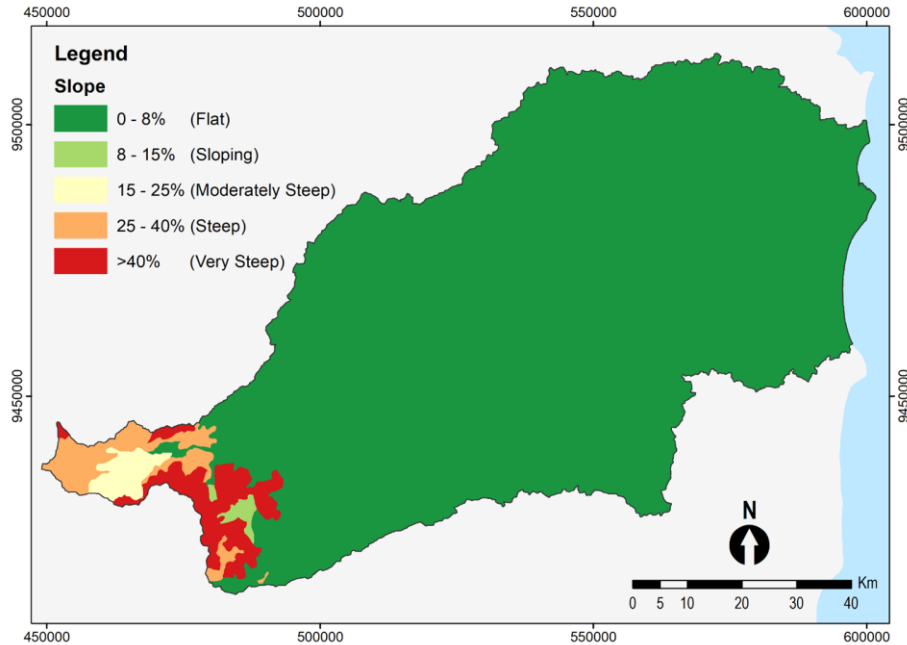


Natural Infiltration Potential

	Total Score			Description
a = 5	3	4	5	Very Low
b = 4	6	7	8	Low
c = 3	9	10	11	Moderate
d = 2	12	13	14	High
e = 1			15	Very High

Source: Minister of Forestry of the Republic of Indonesia Number P.32/MENHUT-II/2009 about Procedures for Preparing a Technical Plan for Forest and Land Rehabilitation of Watershed (RTkRHL-DAS)

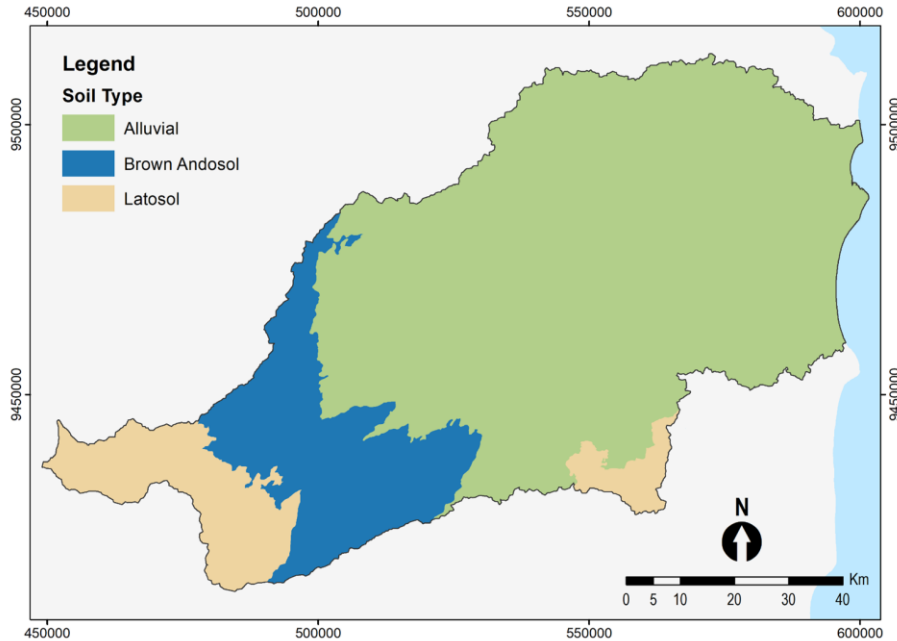
Result & Discussion: Slope



Class	Slope	Description
I	<8	Flat
II	8-15	Sloping
III	15-25	Moderately Steep
IV	25-40	Steep
V	>40	Very steep

- Lower slope = higher infiltration rate
- This map illustrates that Seputih Watershed is **dominated by flat slope**, which means **higher infiltration rates**.
- Higher slope rates are located in the western side of Seputih Watershed because there is Kasih mountain complex consisting of Mount Parus, Mount Tangkit Tebak, and Mount Cakung.

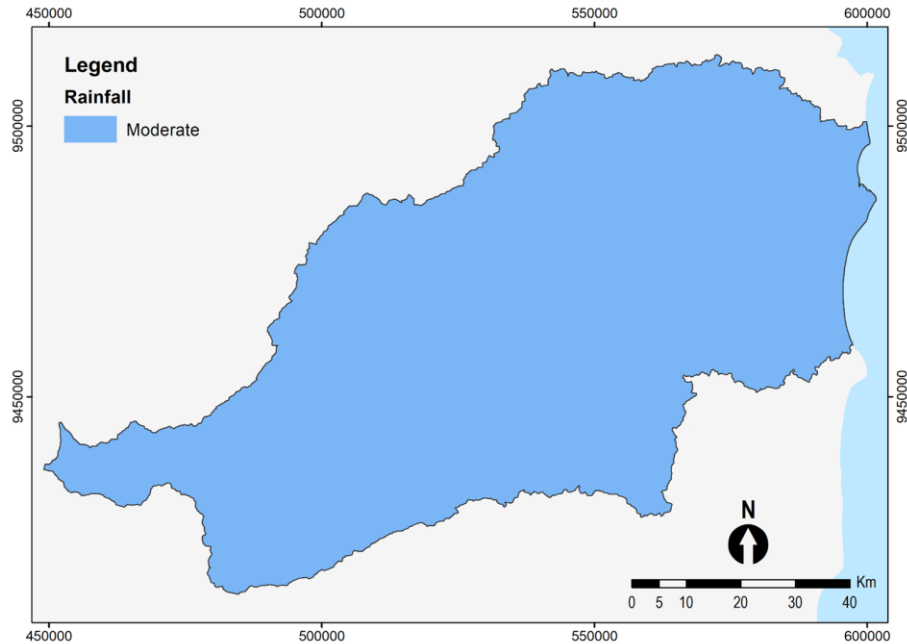
Result & Discussion: Soil Type



Class	Description	Soil class
I	Big	Black andosol
II	Somewhat big	Brown andosol
III	Moderate	Regosol
IV	Somewhat small	Latosol
V	Small	Alluvial

- Soil Types in Seputih Watershed consists of **Alluvial, Brown Andosol, and Latosol** that formed from various geological formation
- Seputih Watershed dominated by **volcanic** and **fluvial** landforms.
- Soils infiltration potential depends on it content, soils with **high sand content** will have a big potential, while the one with **high clay content** will have lower potential of infiltration.

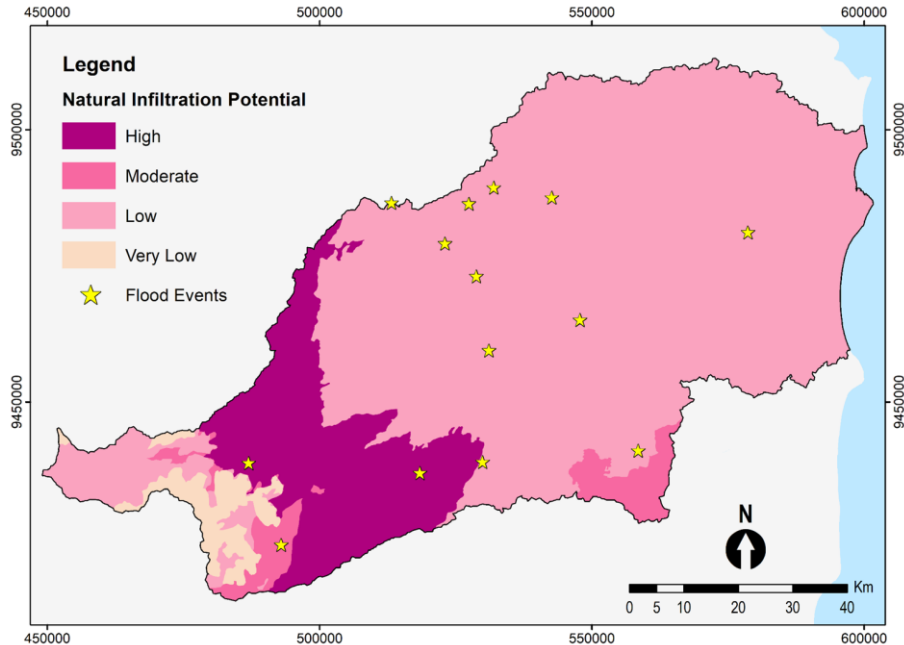
Result & Discussion: Rainfall



Class	Infiltration Rain Classification	Description
I	< 2500	Low
II	2500 - 3500	Moderate
III	3500 - 4500	Somewhat high
IV	4500 - 5500	High
V	> 5500	Very high

- Longer rain period = higher potential infiltration.
- All areas in the Seputih watershed are classified as areas with **moderate rainfall** because they have infiltration rain values of **2500 - 3500 mm** in 2023

Result & Discussion: Natural Infiltration Potential



Class	Total Score	Description
I	3 - 5	Very low
II	6 - 8	Low
III	9 - 11	Moderate
IV	12 - 14	High
V	15	Very high

- Data on **Slope, Soil Type, Rainfall**, which has been given a score for each parameter of 1 to 5, is then overlaid so that it has a total score of 3 to 15.
- Greater total value = greater potential for natural infiltration.
- Natural infiltration potential in Seputih Watershed can be divided into four categories; **high, moderate, low, and very low**
- Area with low natural infiltration potential usually **floods** during the rainy season.

Conclusions

1. The natural infiltration potential in Seputih Watershed are classified into **high, moderate, low, and very low**.
2. This results from this research could give references for the governments to manage the watershed management, such as **sustainable water resource management** by maintaining the balance of surface and groundwater availability. Therefore, the government of Lampung Province can conduct **land use planning**, such as making infiltration wells and reforestation.

References

- Badan Pusat Statistik Lampung Tengah. (2022). *Jumlah Desa/Kelurahan yang Mengalami Bencana Alam Banjir Menurut Kecamatan 2020-2022*. Sumber <https://lampungtengahkab.bps.go.id/>
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- Lal, R. (2000). Integrated Watershed Management in the Global Ecosystem. In *Soil and Water Conservation Society*. CRC Press.